

Self-alignment of active Si bridge using solder joints capillary forces

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Abstract—This paper introduces and validates a novel "Si bridge first" approach to IBM's DBHi packaging technology, which exploits the self-alignment capability of mass reflow (MR) to enable effective assembly of double-sided TSV-type Si bridges onto a substrate via solder joint surface tension. By understanding the relationship between geometry and capillary forces, we demonstrate that high-throughput MR assembly can replace thermocompression bonding (TCB).

Fluidic simulations of vertical and lateral capillary forces as a function of solder thickness between metallized pillars defined optimal configurations for solder-on-pillar interconnections. A semi-additive manufacturing process supported these configurations. A design of experiments evaluated alignment effectiveness on Si bridges assembled on Si substrates. Results using a single 100 μm diameter pillar on each bridge corner demonstrated consistent alignment accuracy under 2 μm , achieving in some cases accuracies better than 1 μm , while mechanical profiling showed z-tilt variations below 1 μm . Comparisons of various corner pillar (1–4 bumps per corner) and full matrix designs confirmed that self-alignment capability is consistent. Controlled lateral offset experiments supported the ability of the configurations to accommodate industry standard flip chip pick and place tolerances.

Thermo-mechanical finite element analysis (FEM) simulations compared TSV-type Si bridges with two-sided bonding to single-sided designs, both on organic substrates in a DBHi module configuration. Results showed no significant effect on Si chip, Si bridge and substrate warpage. Si chip and underfill stresses were also equivalent. However, corner stresses in the Si bridge were significantly reduced in the two-sided bonding design due to the presence of the bottom side solder interconnections. While further experiments are necessary to validate production scaling, this study highlights the potential of MR-based solder self-alignment for TSV-type bridge assemblies with high precision and throughput, positioning it as a viable alternative to TCB.

Keywords— Si bridge, surface tension, solder joints, reflow, heterogeneous integration.

I. INTRODUCTION

As the incremental benefits of transistor scaling decline, there is a growing need for heterogeneously integrated multi-chip packages to enhance bandwidth and minimize data transfer latency. These advanced packages are particularly beneficial for AI workloads that require high-speed communication with low power dissipation per bit. Examples of such packages include silicon (Si) interposers, high-density laminates, 3D, and fan-out packages [1–4]. In a Si interposer package, Si chips are interconnected via redistribution layers mounted on a thinned Si interposer substrate, which is then connected to a laminate substrate using through-silicon vias (TSVs). Spanning across the surface of all chips, this approach can be costly.

To address this cost issue, silicon bridges (Si bridges) [5] have been proposed. Here, small, cost-effective Si substrates are used only where interconnection between chips is required, allowing chips to laminate connections elsewhere. Several Si bridge solutions, such as EMIB [6], EFB [7] and DBHi [8] are available. The latter, developed by IBM, eliminates the need to embed the bridge into the laminate substrate or mold compound, offering a more straightforward assembly process. While the free space between bridges can obviate the need for TSVs, Si bridges comprising TSVs can further increase electrical performance characteristics such as power delivery and off-die parasitics [9]. However, the need to interconnect both sides of such bridges imposes stringent controls on lateral alignment and interconnect height. Thermocompression bonding (TCB) has been favored thus far to achieve this level of control, despite its lower throughput compared to mass reflow (MR) and lack of auto-alignment capability.

In this paper, we argue and demonstrate that MR interconnection and its inherent auto-alignment of solder bumps can indeed be implemented for high throughput mass production of TSV-type Si bridge assemblies. By thoroughly understanding

the interrelationship between geometry and capillary forces, and accordingly designing the solder portion of the Cu pillar, a "bridge first" approach is proposed. Here, the Si bridge is roughly positioned on a substrate yet accurately re-aligned during solder reflow at a predictable target height, thus enabling precise connection of multiple active dies onto the Si bridge's top surface.

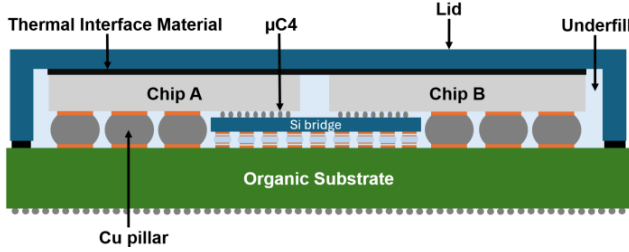


Fig. 1. DBHi Si bridge first module

To validate the feasibility of this method, we begin with fluidic simulations to evaluate the vertical and lateral forces generated by solder confined between two pads, providing insights into interconnect stability and alignment. The simulations propose several metallic pillar configuration designs, which are then evaluated for optimal assembly compatibility. Finally, thermo-mechanical finite element (FEM) simulations based on these geometries are presented that compare interconnection stresses in such TSV-type Si bridges to single-sided designs, to predict mechanical reliability.

II. METHODOLOGY

A. Fluidic Simulations

The self-assembly mechanism directly results from the minimization of the liquid's surface energy. This surface energy, E , is proportional to the liquid's surface area, S , and is expressed by the following equation:

$$E = \gamma S \quad (1)$$

where γ represents the surface tension. It reaches equilibrium when the surface is minimized, *i.e.*, when the component is aligned. Self-alignment is a well-known phenomenon during chip assembly using solder balls [10]. This process relies on capillary forces, F_x , F_y , and F_z , which are expressed as follows:

$$F_x = \frac{\partial E}{\partial x} = \gamma \frac{\partial S}{\partial x} \quad (2.1)$$

$$F_y = \frac{\partial E}{\partial y} = \gamma \frac{\partial S}{\partial y} \quad (2.2)$$

$$F_z = \frac{\partial E}{\partial z} = \gamma \frac{\partial S}{\partial z} \quad (2.3)$$

However, to control this phenomenon, it is essential to quantify these capillary forces involved in the chip-metallized pillar system. The forces involved in a SAC305 liquid soldering process between two metallized pads were simulated using the Surface Evolver software [11]. Knowing the vertical forces generated by the system and the weight of the bridge, the size and number of metallized pillars needed to support the Si bridge and prevent its collapse for a given condition can be determined.

The explored parameters were pad geometry (square or round) and dimensions, vertical spacing between the two pads, and horizontal (x , y) offset between the top and bottom pads. Simulation results were used to propose several Si bridge bottom side designs for assembly validation.

B. Bridge fabrication

To cost effectively emulate TSV type Si bridges, standard Si bridges comprising top side Cu pillar structures were used. To protect these structures from subsequent processing, the Si bridges were first temporarily bonded to a carrier. A semi-additive process was then used to create the bottom side Cu-pillar/solder structures on both the Si bridge and the Si substrate. The fabrication steps and materials used are illustrated and detailed in Fig. 2.

Application requirements dictated a distance of 25 μm between the bottom side of the Si bridge and the matching substrate. Therefore, bottom bridge pillar structures consisted of a 4 μm thick UBM (Cu/Ni/Cu) and 15 μm of Sn, while the remaining height was provided by 4 μm of UBM and 2 μm of Sn on the Si substrate.

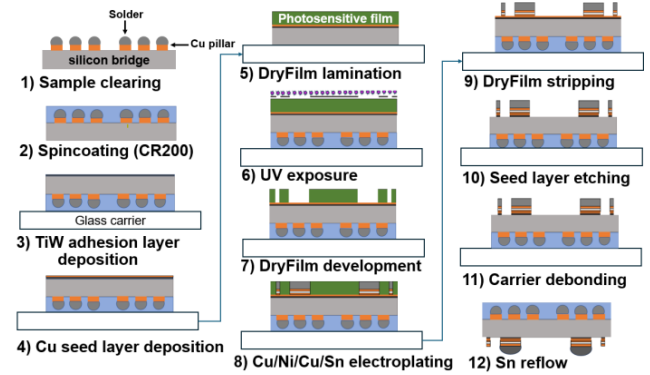


Fig. 2. Metallized pillars under Si bridge process flow

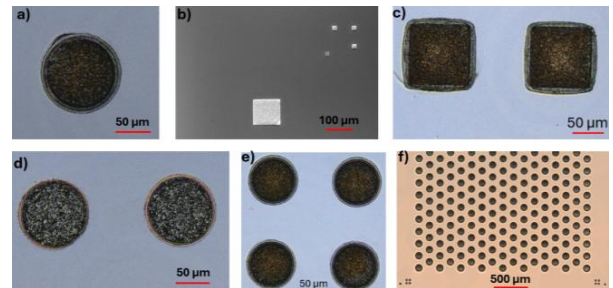


Fig. 3. Different configurations of metallized pillars

The pillar designs used are shown in Fig. 3, with configurations ranging from 1 to 4 pillars per corner of 100 μm width, as well as a full staggered pillar matrix of 80 μm diameter and 150 μm pitch. Both square and circular designs were examined.

C. Si bridge assembly process and characterizations

The assembly process of the Si bridges consisted of three steps, as shown in Fig. 4.

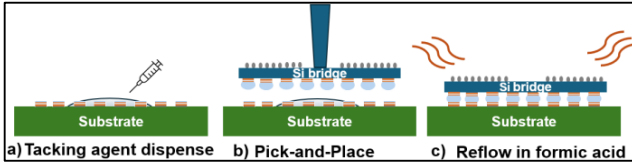


Fig. 4. Si bridge assembly process

A tacking agent was applied to the Si substrate. The Si bridge was placed in rough alignment onto the substrate using pick-and-place (PnP) equipment. Finally, the assembly was reflowed in a furnace with a formic acid de-oxidizing atmosphere.

The assembly process was tested with bridges comprising either a single pillar at each corner or a full matrix. Self-alignment performance was assessed by measuring lateral misalignment. For this purpose, metallic markers were used, fabricated simultaneously with the metallized pillars, as shown in Fig. 5.

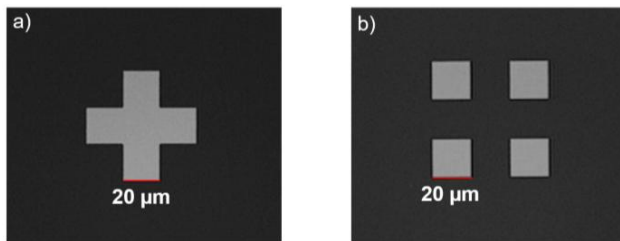


Fig. 5. a) Metallic marker under Si bridge. b) Metallic marker on substrate surface

Corresponding alignment squares were also created on the substrate surface, in the areas designated to receive the Si bridge. Ideal lateral alignment would be achieved when the cross under the Si bridge is perfectly centered relative to the squares on the Si substrate along both the x and y axes.

To measure lateral alignment, the optical transparency of infrared light through the Si bridge was exploited to observe the alignment markers using infrared microscopy. Centers of the cross and the squares were determined, and the displacements along the x and y axes were measured. Given the limitations of infrared microscopy in terms of resolution, this characterization was complemented with scanning electron microscopy (SEM) measurements.

D. Finite Element Modeling

Thermo-mechanical analyses by finite element method (FEM) were conducted to understand the impact of the two-sided bonding Si bridge design on interconnection stresses as compared to single-sided bonding of a standard Si bridge.

The analysis model of the DBHi module with two-sided bonding Si bridge is shown in Fig. 6.

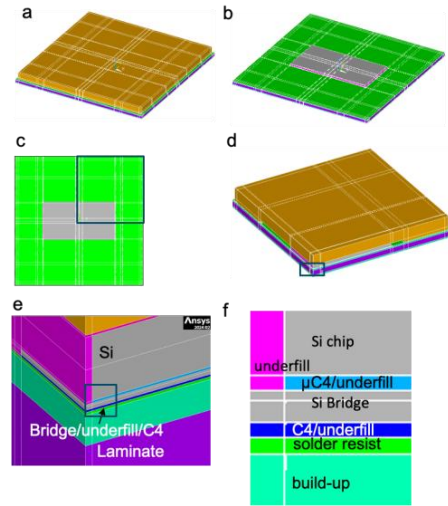


Fig. 6. Macro model of DBHi module: a) an inclined view of the model. b) the model without the lid. c) Position of the 1/4 model. d) 1/4 model. e) Enlarged view of the 1/4 model. f) Side view showing the layers near the Si bridge.

The model consists of an organic laminate, underfill layers with solder joints, two Si chips and a Si bridge connecting the chips. A sub-modeling technique was used to evaluate the strain in the solder materials for the solder joints. Micro models for the solder joints regions are shown in Fig. 7.

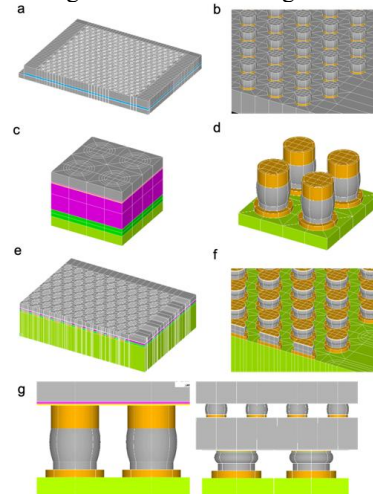


Fig. 7. Micro models of DBHi module: a) μC4 region. b) μC4 bumps. c) Corner C4. d) C4 bumps. e) Solder bump region under the Si bridge. f) Solder bumps under the Si bridge. g) Side views of the models

Since the module is quadrant symmetric, a 1/4 model was adopted. The layer underneath the Si bridge is underfill for the DBHi module with single-sided bonding Si bridge and is a composite layer comprising an equivalent property of underfill and solder joints for the module with two-sided bonding Si bridge. Equivalent material properties were also applied to the joint layer between the Si chip and laminate, to the layer between the Si chip and Si bridge and to the circuitry layers in the laminate. Plastic properties were assumed for the copper lid. Temperature dependent linear properties were used for the laminate core and the build-up material. The Anand model was used for the visco-plastic property of the solder material in the C4, μ C4 and the joints for the self-alignment. Linear properties were used for the Si chip and Si bridge.

Temperature loading condition for the analyses was based on cooling down from 150°C to -40°C and one cycle of thermal cycling condition to estimate the risk during the assembly and subsequent thermal cycle testing. A stress-free temperature of 150°C was used for the whole package.

The effects of a two-sided Si bridge on the following metrics were evaluated: Si bridge warpage, the stress on the Si chip and Si bridge, the stress on the underfill/solder joint layers, and the inelastic strain range for solder materials in the solder joints.

III. RESULTS

A. Fluidic simulations

Simulations provided vertical and horizontal capillary forces F_{cap} , as a function of the initial solder volume, leading to the identification of three response cases for our study, as shown in Fig. 8.

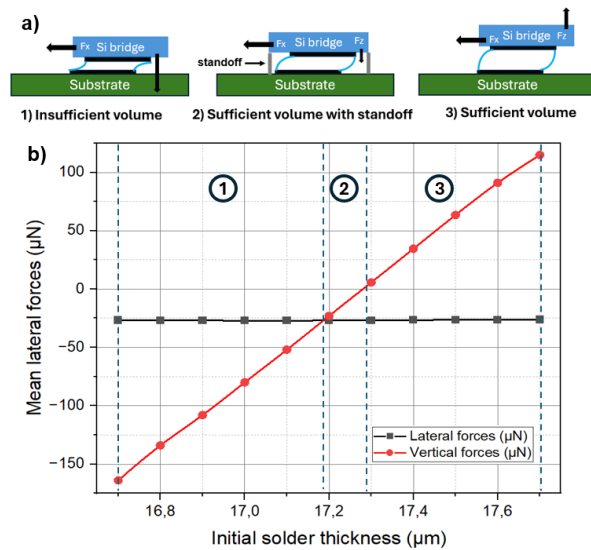


Fig. 8(a) Possible response cases during self-alignment with (b) corresponding graphical representation using Surface Evolver

The first case occurs when the initial solder volume is insufficient, resulting in gravitational forces F_g being too

important compared to F_{cap} , preventing the motion of the Si bridge.

In the second case, when the curves intersect, the vertical F_{cap} exceeds the lateral F_{cap} . However, the vertical F_{cap} remains smaller than F_g , implying a collapse of the Si bridge. In such cases, standoffs, which are mechanical supports that partially hold the Si bridge to prevent it from collapsing, can be used. During lateral self-alignment, frictional forces $F_{friction}$, between the Si bridge and the standoff will occur. If $F_{friction}$ is weaker than lateral F_{cap} , the self-alignment can still happen.

Finally, in the third case, when the vertical forces turn positive, F_{cap} is stronger than F_g and the Si bridge does not collapse. Then, self-alignment becomes possible without the need for standoffs to maintain the target height.

Calculations of these capillary forces enabled the determination, for a given diameter, of the number of pads required for initial support and for self-positioning of the Si bridge. Results are summarized in Table 1.

TABLE I. SMALLEST NUMBER OF PILLARS PER CORNER

Pad size (μ m)	Vertical force (μ N)	Minimal number of corner pads
100	27.01	1
75	19.59	1
50	12.58	1
25	6.03	1
15	4.61	2

Simulations carried out to compare square and round geometries for equivalent configurations revealed no significant differences in alignment response.

B. Si bridge assembly and characterization

Assemblies in this study used Si substrates as organic substrates designs were not available at the time of this publication. Considering the small size of the Si bridges, alignment results are expected to be similar despite the thermal expansion coefficient (CTE) differences between these substrates. Initial tests focused on configurations with one, two, and four pillars per corner to validate the accuracy of previously conducted capillary force simulations.

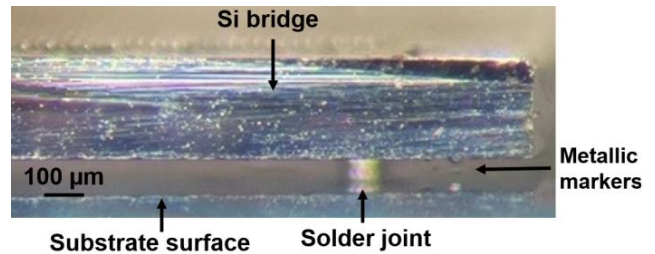


Fig. 9. Self-aligned Si bridge with one corner pillar

Fig. 9 shows an optical image of a self-aligned Si bridge with a single 100 μ m diameter pillar per corner. The bridge has not collapsed, validating the simulation results of Table 1.

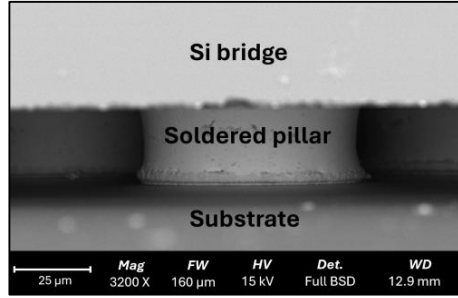


Fig. 10. SEM side view of soldered pillar

Fig. 10 shows a SEM side view image following the assembly process described in Fig. 4. Soldering between the bridge pillar and the substrate was observed, highlighting the effectiveness of the assembly method in maintaining structural stability.

Once the soldering of the Si bridge had been demonstrated, the focus was placed on quantifying the lateral quality of the self-alignment.

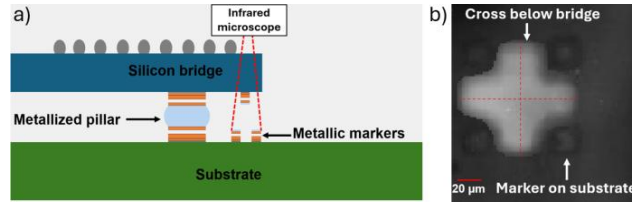


Fig. 11. a) Schematic of top view IR microscopy through Si bridge. b) IR microscopy image of metallic markers

Top view measurements using infrared microscopy facilitated a rapid characterization of the Si bridges, enabling the determination of whether the self-alignment was successful. Fig. 11 illustrates a cross marker precisely aligned with the center of the squares. However, to quantify alignment with single micron accuracy, SEM side view cross-sectional measurement (Fig. 12) was necessary.

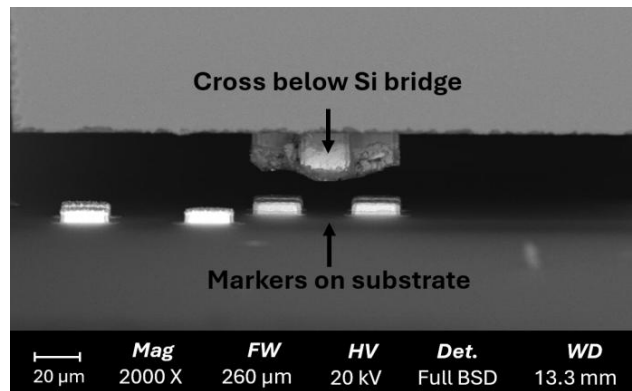


Fig. 12. SEM side view of alignment markers after Si bridge assembly

The misalignments of Si bridges with 1 to 4 pillars per corner were first measured. In all instances, the lateral misalignment

remained below 2 µm, with some cases exhibiting values below 1 µm.

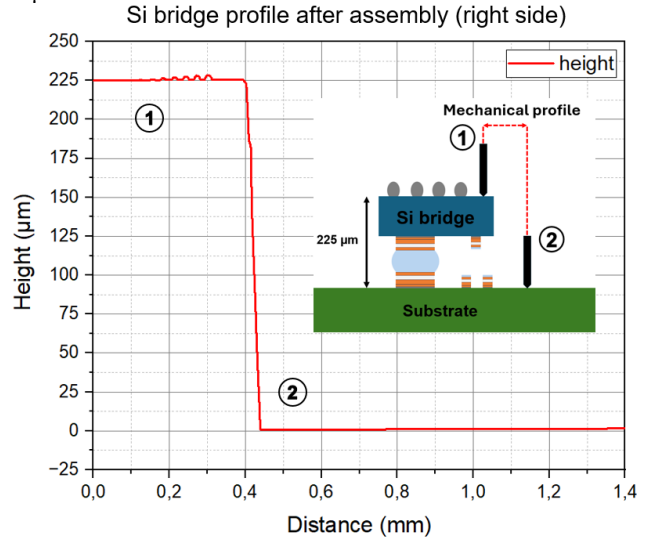


Fig. 13. Si bridge profile after assembly (right side)

After assembly, profilometry was used to measure the Si bridge top surface height relative to the substrate surface at the four edges (left, right, top, bottom), as depicted in Fig. 13. In each case, a z-tilt below 1 µm was observed, confirming the uniformity of thickness achieved during the electrodeposition steps.

C. Limit of self-alignment

Once the proof of concept was established, we proceeded with the assembly of Si bridges using a full matrix pillar design, which would be necessary to enable TSV integration in Si bridges for future applications. Intentionally, x and y offsets were introduced during the placement of the Si bridges to test the self-alignment capabilities under varied conditions.

TABLE II. SELF-ALIGNMENT EFFICIENCY VS OFFSET

Number of Si bridges assembled	Intentional X-Y offset value (µm)	Number of aligned bridges ($\leq 2 \mu\text{m}$)
20	2	20
20	10	17
20	20	14
20	30	4
20	50	0

Results as a function of the initially introduced misalignment are shown in Table II for 80 µm diameter pillars at 150 µm pitch. For offsets up to 20 µm, self-alignment was relatively effective, considering possible errors in initial offset accuracy. However, at 30 µm, the self-alignment capability was very low. This corresponds to an initial offset of 37.5%, approaching the physical limit of capillary forces [12].

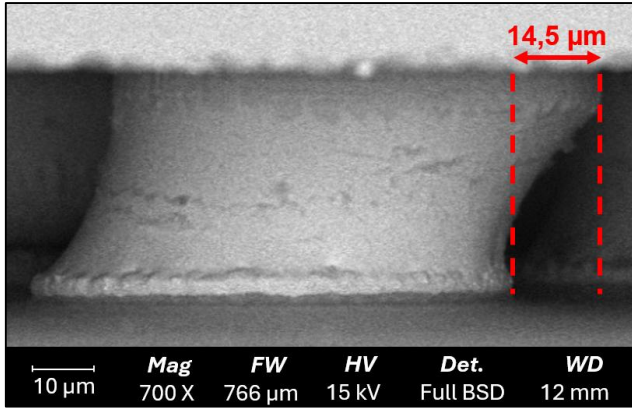


Fig. 14. SEM image of misaligned pillar

A case with an initial offset of 50 μm is shown in Fig. 14. It can be observed that, while effective soldering occurred, a final lateral misalignment of 14.5 μm remained. This highlighted both the action of capillary forces and their limitations.

D. Finite Element Modeling

The results obtained through Finite Element Modeling (FEM) are presented in this section by comparing the two-sided Si bridge configuration to the DBHi module with a single-sided Si bridge. All the results are indicated relative to that of the module with single-sided bonding Si bridge.

Warpage is defined to be the difference in vertical displacement between the center and corner of laminates or Si chips. Comparison of warpage of the laminate and the Si chips are shown in Fig. 15. No significant differences in warpage of any of these components were observed between the two configurations.

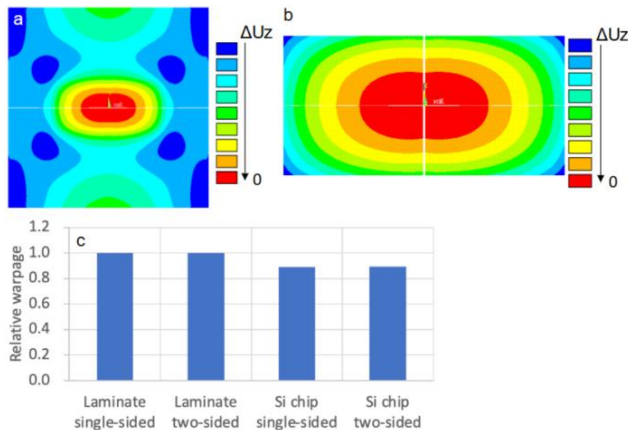


Fig. 15. Contour of the z-displacement for: a) the laminate and b) Si chip for DBHi module with two-sided Si bridge. c) Plot of relative warpage.

Contour of the vertical displacement of the double-sided Si bridge is shown in Fig. 16. Maximum vertical displacement was at the gap between the two chips for both the double-sided and single-sided bonding Si bridge. Vertical displacement is plotted

in Fig. 16b with that of the single-sided bonding bridge. The displacement curves were almost identical.

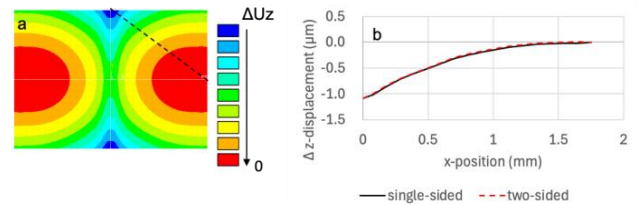


Fig. 16. a) Contour of the z-displacement for Si bridge for DBHi module with two-sided bonding bridge. b) Plot of the z-displacement for the bridge along the dotted line.

Stresses on the Si chips and the Si bridge are compared in Fig. 17. The maximums of the first principal stress are located at the chip corner and at the bridge corner. No significant differences were observed for the Si chip stress. However, the corner stress for the Si bridge was significantly lower for the module with the two-sided bonding compared to the one with the single-sided bonding, as the solder bump matrix provides additional support to the Si bridge. This lower stress could potentially reduce the risk of bridge cracking or underfill delamination.

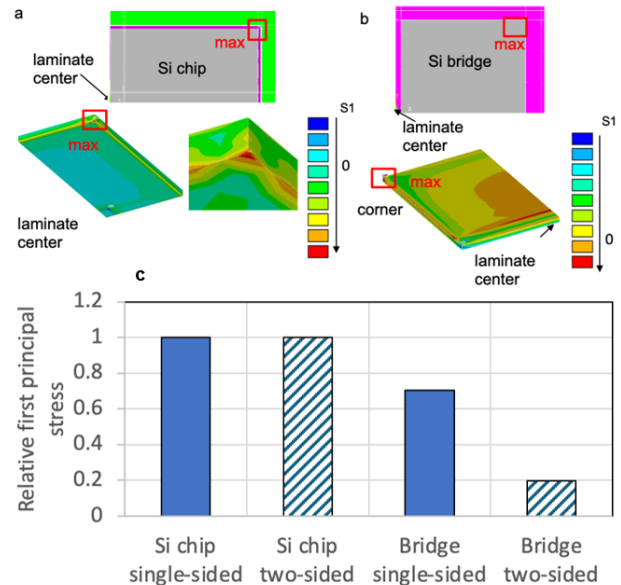


Fig. 17. Stress on Si chips: a) Stress contour for Si chip. b) Stress contour for the two-sided bonding Si bridge. c) Comparison of stresses on Si chip and bridge.

Underfill stresses are compared in Fig. 18. The stress distribution maps revealed that the maximum stress concentrations primarily occurred at the interfaces between the underfill and the C4 and μC4 interconnection structures. Simulations showed that the two-sided configuration did not result in an increase in underfill stress.

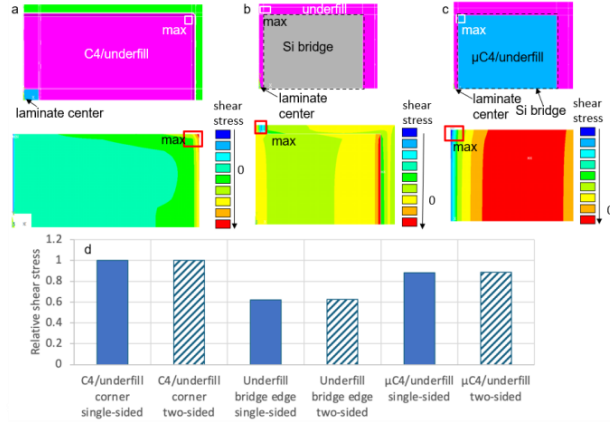


Fig. 18. Comparison of stress on underfill and underfill/solder joints: a) contour of C4/underfill composite layer. b) Solder joints under the Si bridge. c) μ C4/underfill joints. d) Plot comparing stresses on underfill layers.

Strain in solder joints are compared in Fig. 19. Accumulated inelastic strain for solder joints used in self-alignment showed maximum strain under the bridge edge near the Si chip gap. The μ C4 joints also showed maximum strain near the Si chip gap. Inelastic strain range is compared in Fig. 19.e. It shows that the strain range for C4 and μ C4 joints is comparable for the modules with single-sided and two-sided bonding bridge. The inelastic strain range of the solder joints for the self-alignment is less than that of the C4 bumps, suggesting a lower risk of failure.

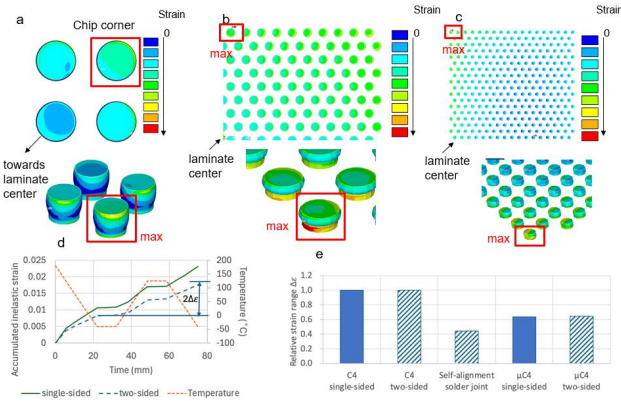


Fig. 19. Strain for solder joints: contour of solder joints for a) C4 bump. b) Solder joints for self-alignment. c) μ C4 bump. d) Plot of time evolution of accumulated inelastic strain, for C4 and solder joints for self-alignment. e) Comparison of relative inelastic strain range

IV. CONCLUSION

This study explored the potential to exploit mass reflow and its auto-alignment capability for cost effectively assembling Si bridges. Through fluidic simulations and subsequent assembly validations, we demonstrated that an appropriate volumetric design and a correspondingly robust pillar fabrication process can achieve lateral auto-alignment below $2\ \mu\text{m}$, and in some cases below $1\ \mu\text{m}$, all the while maintaining a z-tilt below $1\ \mu\text{m}$. We further demonstrated that this can be maintained at initial offsets as high as $20\ \mu\text{m}$ for an $80\ \mu\text{m}$ diameter pillar, thus demonstrating at least 25% auto-alignment and projecting compatibility with industry standard pick and place alignment tolerances.

The thermo-mechanical FEM simulations indicated that the two-sided bonding Si bridge design does not noticeably affect warpage or general stress distribution compared to the single-sided approach. However, the presence of solder interconnections on the bottom side of this two-sided bridge configuration appears to contribute to a lower corner stress on the Si bridge that may result in improved bridge integrity.

These promising preliminary results propose further study regarding production scale of this process, considering inherent challenges such as the handling of a thin, double sided bumped die and the manufacturing consistency of pillar bump volume. Finally, reliability testing of assemblies comprising organic substrates should be carried out to validate both the auto-alignment capability observed for Si substrates and the thermo-mechanical stability proposed by our FEM analysis of double-sided bridge assembly on organic substrates.

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